NATIONAL TECHNICAL RESEARCH ORGANISATION (NTRO)

FLOODGUARD: DAM-BREAK INUNDATION MODELLING SYSTEM

Smart India Hackathon 2026 | Disaster Management Command Platform

Simulation ID: TEST_SIM_001	**Generated:** 2026-10-11 09:48:09 UTC
Study Basin: Periyar River Basin, Kerala	**Dam Facility:** Mullaperiyar Dam
Model Engine: Prototype 2D Flood Propagation Engine	**Risk Severity:** CRITICAL

1. Executive Summary

This report presents hydrodynamic dam-break inundation modeling for **Mullaperiyar Dam** along the downstream reach of the Periyar River. A simulated breach of width **85.0 m** initiates a peak discharge of **25,652 m³/s**, causing total flood inundation across an estimated **198.84 km²** of downstream river valley terrain. Flood arrival at the first major downstream habitation occurs within **15.0 minutes**. The incident is categorized as **CRITICAL** requiring prompt evacuation dispatch.

2. Primary Hydrological & Impact Metrics

Metric Description	Computed Value	Baseline / Operational Significance
Peak Breach Discharge (Qp)	25,652 m³/s	Broad-crested weir formulation & Froehlich verification
Maximum Inundation Depth	7.9 meters	Directly downstream of dam breach invert
Total Flood Extent	198.8 km²	Derived from 2D DEM cell depth matrix (h > 0.05m)
Minimum Arrival Time	15 minutes	Wave celerity along primary river corridor
Affected Habitations	6 settlements	Spatial intersection with census village boundaries
Population at Direct Risk	16,932 people	Downstream inhabitants within flood extent
Inundated Roads	18.6 km	Transportation corridors requiring traffic shutdown
Critical Facilities Impacted	1 Bridges, 1 Hospitals	Emergency services requiring relocation

3. Hydraulic Breach Formulation

Outflow discharge was calculated using hydraulic weir formulation: $Q(t) \approx C \cdot B(t) \cdot [H(t)]^{1.5}$, where discharge coefficient C = 1.7, breach width B = 85.0 m, and breach formation time = 1.2 hours. Initial water level was modeled at 108.0 m above datum with Manning's roughness n = 0.035.

4. Downstream Settlements & Infrastructure Breakdown

Settlement Name	Type	Population	Status	Est. Arrival
Upputhara Panchayath	Panchayath	6,320	FLOODED	T+55 min
Ayyappancoil Settlement	Village	3,940	FLOODED	T+55 min

Chapathu Village	Village	1,850	FLOODED	T+55 min
Marykulam Ward	Ward	2,400	FLOODED	T+55 min
Karimkunnam Downstream	Settlement	4,120	THREATENED	T+45 min

5. Modeling Assumptions & Limitations (SIH Scientific Integrity)

1. **Hydrodynamic Solver Attribution**: Results above are generated by the *Prototype 2D Flood Propagation Engine*. Delft3D and Smooth Particle Hydrodynamics (SPH) modular adapters are connected via standardized pipeline schemas.
2. **Terrain Resolution**: Based on 30m resolution digital elevation dataset with local hydrological conditioning.
3. **Decision Support Notice**: This platform provides automated decision support and simulation analysis; all disaster management actions should follow official state and national disaster management authorities (NDMA/SDMA) SOPs.